Planetary Rover Power Management Simulator

COMP0068 MIPS32 Coursework 2026/27

Rationale and Learning Objectives

This coursework is designed to move you beyond the basics of MIPS assembly and immerse you in the low-level interaction between computation and user-interaction.

You will design a MIPS32 planetary Rover Power Controller (RPC) and render its power budget in real-time. This task cannot be solved by copying found code; it requires original architectural design and problem-solving. Successfully completing this project will demonstrate a masterful grasp of the MIPS32 ISA (Instruction Set Architecture).

Upon completion, you will be able to:

- Design and implement a complex program in MIPS.
- Implement algorithms using arithmetic and bitwise operations.
- Implement procedures and branching.
- Handle real-time user input within a simulation loop.
- Structure and comment code to professional standards.

Project Specification

A world-leading space robotics company recruits your group to develop a MIPS32 application. The program implements a Rover Power Controller (RPC). A mission operator interacts with the RPC through a set of commands. The rover can be recharged, solar-aligned, docked, tasked with a traverse, ... and reset.

The list of commands is as follows:

1. Recharge
2. Solar align
3. Dock
4. Traverse
5. Reset
6. Quit

Each command will affect the battery charge level of the rover differently. Table 1 gives detailed information about each command.

Note: The rover's battery charge decreases naturally with time, even if there is no interaction from the operator, because the onboard heaters and telemetry radio draw power continuously. To avoid reaching a charge level of 0 (the rover goes offline), the operator should recharge, dock, ... the rover.

Table 1 — Command set

	Command name	Command shortcut	Effect on charge level	Example
1	Recharge	`C n`	Increase charge level by n units	`C 3` increases the charge level by 3 units
2	Solar align	`S n`	Increase charge level by 2 x n units	`S 1` increases the charge level by 2 x 1 = 2
3	Dock	`D n`	Increase charge level by 2 x n units	`D 2` will increase the charge level by 2 x 2 = 4
4	Traverse	`T n`	Decrease charge level by 3 x n units	`T 2` decreases the charge level by 3 x 2 = 6
5	Reset	`R`	Reset RPC to its original state (operator-defined IBC (Initial Battery Charge))	Charge level resets to IBC
6	Quit	`Q`		Quits the application

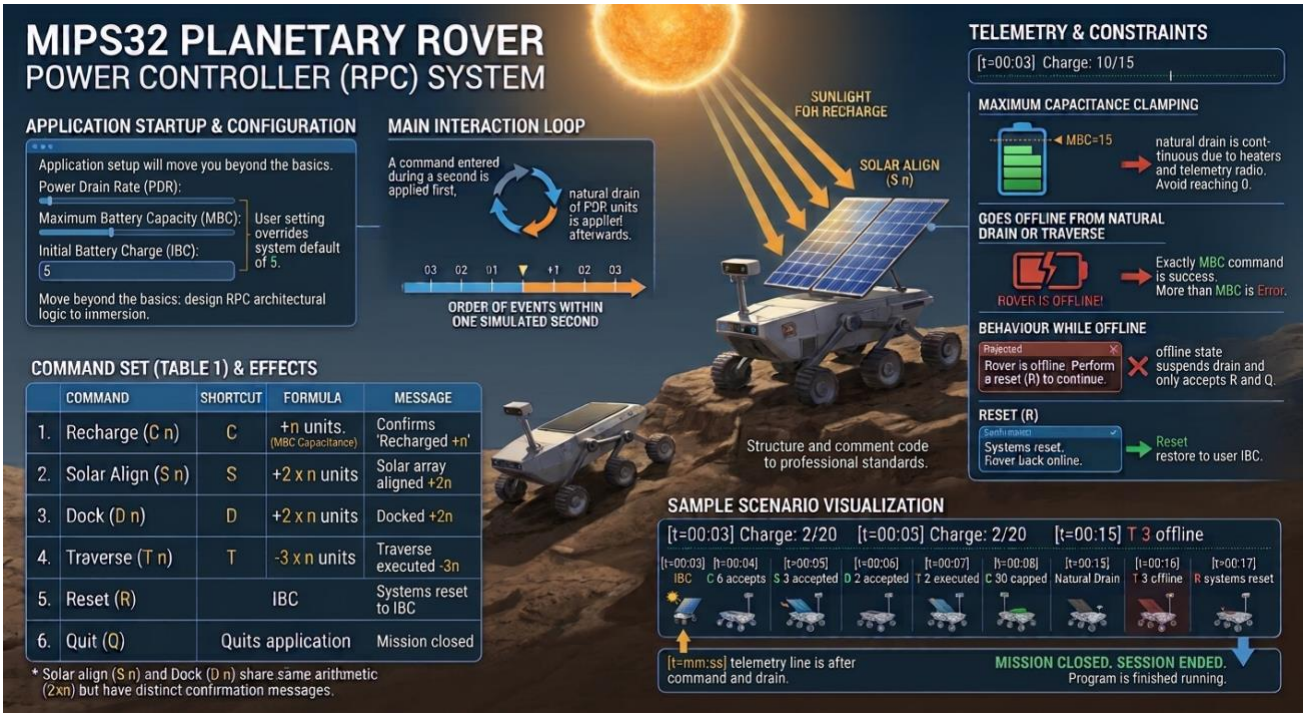


Figure 1: MIP32 Planetary Rover Power Controller System (for illustrative purposes only, your system should follow scenario below)

Note on commands 2 and 3: *Solar align* means the rover stays roughly where it is and orients its solar panels, to face the Sun as directly as possible to recharge its battery. *Dock* means the rover drives to a fixed station, such as a lander or base and physically connects or aligns with it. That connection can provide charging from the station's larger power source.

They are distinct commands with distinct shortcuts, distinct confirmation messages and distinct operational meanings, but identical arithmetic — your dispatch logic must recognise both letters and print the correct message for each, even though the value they compute is the same.

Requirements

- Upon startup, the application should allow the operator to set three parameters:
  - the natural power drain rate PDR. If not set, default value is 1 unit of charge every 1 second.
  - the maximum battery capacity MBC. If not set, default value is 15.
  - the initial battery charge, IBC. If not set, default value is 5.
- The charge level of the rover increases and decreases based on the operator's input (see Table 1).
- The commands should be entered in capital letters: C, D, etc. (as shown in Table 1).
- If the rover reaches a charge level of 0, it goes offline. To be brought back online, a reset should be performed.
- A reset can be applied at any time of the mission life cycle of the rover.
- If a command results in increasing the charge level to more than the maximum, the real increase is capped to the maximum.
  - Example: if the current charge level is 14 and MBC = 15. If the operator enters C 3, the level should be capped to the maximum 15 (MBC) and the following error message should be displayed:

```
Error, maximum battery capacity reached! Capped to the Max.
```

- Same for commands that decrease the charge level, such as T.
  - Example: if the current charge level is 2 and the operator types T 3, the rover goes offline (charge level 0). A corresponding error message is displayed:

```
Error, charge level equal to or less than 0. Rover is offline!
```

Clarifications

The following rules remove ambiguity in the simulation loop. Your implementation must follow them, because the sample scenario below depends on them.

- **Order of events within one simulated second.** A command entered during a second is applied *first*, and the natural drain of PDR units is applied *afterwards*. The charge shown on the `[t=mm:ss]` telemetry line is the value after both steps.
- **Clamping.** The charge level is always clamped to the range 0 to MBC inclusive. It is never displayed as negative and never displayed above MBC.

- **Exactly reaching the maximum.** The capping error message is displayed only when the requested increase would have exceeded MBC. A command that lands exactly on MBC is a normal success and prints its usual confirmation.
- **Going offline through natural drain.** The rover can go offline from the drain alone, with no command entered. When it does, the same offline error message is displayed.
- **Behaviour while offline.** Once the rover is offline, the natural drain is suspended and only `R` and `Q` are accepted. Any other command is rejected with:

```
Rover is offline. Perform a reset (R) to continue.
```

- **Reset.** `R` restores the charge to the operator-defined IBC not the default one 5.
- **Invalid input.** `n` must be a positive integer. An unrecognised command letter, a missing argument, or a non-positive `n` is rejected without changing the charge level:

```
Invalid command. Use C n | S n | D n | T n | R | Q.
n should be positive
```

Sample scenario

1. Application startup and initial configuration

Upon assembling and running the program in a MARS simulator, the system boots and prompts the operator to configure the mission parameters. The operator can press Enter to accept the default values.

```
=== ROVER POWER CONTROLLER v1.0 ===
Enter power drain rate PDR (units/sec) [default 1]: 2
Enter maximum battery capacity MBC [default 15]: 20
Enter initial battery charge IBC [default 5]: 8

Configuration accepted. PDR = 2, MBC = 20, IBC = 8
Rover online. Beginning telemetry.
```

Figure 2

2. Main interaction loop and natural drain

The program now enters the main loop. A timer simulates the passage of mission time, and the charge level decreases by the PDR (2 units) every simulated second. The clock starts ticking immediately, so by the time the operator reads the display three seconds have elapsed and the charge has fallen from the IBC of 8 to 2. The operator is presented with a command prompt, which now includes the `Q` option.

```
[t=00:03] Charge: 2/20
Command (C n | S n | D n | T n | R | Q):
```

Figure 3

3. Command: Recharge (`C n`)

The operator decides to recharge the rover to counteract the natural drain.

```
Command (C n | S n | D n | T n | R | Q): C 6
Recharge accepted: +6 units.
[t=00:04] Charge: 6/20
```

Figure 4

4. Commands: Solar align (`S n`) and Dock (`D n`)

The operator uses two different commands that have the same mathematical effect (2 x n) but different confirmation messages. `S` tilts the solar array toward the sun; `D` couples the rover to a wayside charging post. Both deliver 2 units of charge per unit of `n`.

```
Command (C n | S n | D n | T n | R | Q): S 3
Solar array aligned: +6 units.
[t=00:05] Charge: 10/20

Command (C n | S n | D n | T n | R | Q): D 2
Docked at charging station: +4 units.
[t=00:06] Charge: 12/20
```

Figure 5

5. Command: Traverse (`T n`)

The operator sends the rover across rough terrain, which has a strong negative effect.

```
Command (C n | S n | D n | T n | R | Q): T 2
Traverse executed: -6 units.
[t=00:07] Charge: 4/20
```

Figure 6

6. Testing constraint: capping at maximum capacity (MBC)

The operator tries to push a large amount of charge into the battery. The charge should be capped at the MBC (20). Note the ordering: 4 + 30 = 34 is capped to 20, and the drain of 2 units for that second is then applied, so the telemetry line reads 18.

```
Command (C n | S n | D n | T n | R | Q): C 30
Error, maximum battery capacity reached! Capped to the Max.
[t=00:08] Charge: 18/20
```

Figure 7

7. Testing constraint: charge depletion to zero (offline)

Seven seconds pass with no command, taking the charge from 18 down to 4. The operator then makes a crucial mistake.

```
[t=00:15] Charge: 4/20
Command (C n | S n | D n | T n | R | Q): T 3
Error, charge level equal to or less than 0. Rover is offline!
[t=00:16] Charge: 0/20  -- ROVER OFFLINE --
```

Figure 8

8. Command: Reset (R)

The operator brings the rover back online using the Reset command. The rover returns to the operator-defined initial state (IBC of 8), not the system default of 5. The drain was suspended while the rover was offline and resumes on the following second, so this telemetry line shows exactly 8.

```
Command (C n | S n | D n | T n | R | Q): R
Systems reset. Rover back online.
[t=00:17] Charge: 8/20
```

Figure 9

9. Command: Quit (Q)

The operator decides to end the simulation session gracefully using the Quit command.

```
Command (C n | S n | D n | T n | R | Q): Q
Telemetry link closed. Mission session ended.
-- program is finished running --
```

Figure 10

Note: Your implementation must replicate the output exactly as demonstrated. Beyond that, you have the freedom to innovate and add new features.

Use of Generative AI

This assessment falls within category 2 of the UCL "Use of GenAI in Assessment"; Category 2 means GenAI tools can be used in an assistive role. More details can be found on the UCL assessment guidance pages.

Marking

The maximum total is 20 marks. The breakdown of the marks is as follows:

Criterion	Marks
Correct implementation of the different commands (`C`, `S`, ..., `Q`)	12
Display the informative text lines exactly as per the sample output above	1
A text file `disclaimer.txt` with the content described above	1
Any extra feature that adds value to the application[1]	6
Total	**20**

[1] For those who decide to implement extra features: view this prototype as your opportunity to impress a company board. To truly stand out, we encourage you to build upon the foundation with creative features that add significant value. This is your chance to prove you can innovate and think beyond the checklist. The core functionality should behave exactly as described earlier though.

Group work and individual mark

The final mark of each group member is devised by considering the individual contribution to the project (using IPAC scores). See the IPAC methodology presentation on Moodle.

All members of the group should contribute to the implementation of the project. Poor engagement with the project will decrease your individual mark.

It is highly recommended that you select a project leader who will organise the group and supervise the allocation of tasks and the progress of the implementation.

Deliverables

As an individual group member:

1. Each member of the group is required to submit an IPAC form on Moodle *by the deadline*. Failing to do so, will reduce your individual mark.

As a group:

2. You should submit one **.asm* file containing your solution, using the link on Moodle. Email submissions are not accepted.
3. A text file *disclaimer.txt* detailing:
   (a) whether you used Gen AI and to what extent, and,
   (b) any extra feature you have implemented.

Plagiarism

UCL enforces a firm policy regarding plagiarised content. Any attempt to include existing code (from former students' work or other sources) into your application "could be penalised for Academic Misconduct, which is defined as any action or attempted action that may result in you obtaining an unfair academic advantage."

Refer to the UCL academic integrity page.

End.